Xirui Li

4244679380 | xiruili@umd.edu | xirui-li.github.io | xirui-li | xirui-li | Scholar

Research Summary

CS Ph.D. student researching **how digital agents represent and interact with the physical world**. Developing multimodal agents for editable 3D scene reconstruction and embodied manipulation, supported by scalable agentic training and evaluation methods for vision-language models. Previously studied emergent visual reasoning and the effects of RL on multimodal models. Applied Research Intern with AWS AgentCore.

Education

- **University of Maryland, College Park** — *2025 – Present*
  Ph.D. in Computer Science | Advisor: Prof. Tianyi Zhou — College Park, USA
- **University of California, Los Angeles** — *2022 – 2024*
  M.S. in Electrical Engineering | Advisor: Prof. Cho-Jui Hsieh — Los Angeles, USA
- **Technical University of Munich** — *2018 – 2022*
  B.S. in Electrical Engineering | Advisor: Dr. Hao Shen — Munich, Germany

Current Research

- **How do digital agents represent the physical world?** In **LEGO-Anything**, developing coding agents that reconstruct editable 3D scenes, using code as a structured representation of the physical world.
- **How do digital agents interact with the physical world?** In **Guava**, developing an effective, universal harness for embodied manipulation, enabling digital agents to act in and interact with the physical world.

Selected Publications & Research

- **LEGO-Anything: Coding Agents for 3D Scene Reconstruction** [ArXiv 2026] [Project]
  First work evaluating and enhancing coding agents for 3D reconstruction; Hugging Face Daily Paper: 138 likes.
  Developed an **Image-to-Code framework** that iteratively writes, executes, and revises Blender programs to reconstruct editable 3D scenes from single images. Introduced LEGO-Bench (208 images, 104 scenes) and a training-free harness plugin that improved all six evaluated models, with up to **62.7% relative gains**.
- **Guava: Distilling Frontier VLMs into a Compact Agent through a Robotic Manipulation Harness** [ArXiv 2026] [Code]
  First work to distill frontier VLMs' manipulation capabilities into a compact agent.
  Designed a robotic manipulation harness combining **perception–reasoning–action loops**, semantic actions, and multimodal observations. Distilled frontier-model capabilities into a **4B agent using fewer than 2K simulated trajectories**, achieving comparable performance in simulation and real-world experiments.
- **What does RL improve for Visual Reasoning? A Frankenstein-Style Analysis** [COLM 2026] [Code]
  The first work to disentangle capability gains in multimodal post-training.
  Developed a causal analysis framework using component swaps, parameter comparisons, and model merging to localize RL improvements. Found that **mid-to-late transformer updates improve vision-to-reasoning alignment**, with transferable gains rather than uniform improvements in visual perception.
- **VisualThinker: R1-Zero's "Aha Moment" in Visual Reasoning** [ICLR 2026 WS] [Code]
  The first replication of DeepSeek-R1-style emergent reasoning in a vision-language model; 600+ GitHub stars.
  Applied RL directly to Qwen2-VL-2B without an SFT cold start, eliciting **self-reflection and emergent multimodal reasoning**. Achieved **59.47% accuracy on CVBench**; analyzed why instruct-model initialization and naive length rewards fail to induce similar reasoning behavior.
- **ClawEnvKit: Automatic Environment Generation for Claw-Like Agents** [ArXiv 2026] [Code]
  The first work to propose bespoke environment synthesis for claw-like agents.
  Built a **parse–generate–validate pipeline** that creates task specifications, tool interfaces, and scoring rules from natural language. Generated Auto-ClawEval with **1,040 environments across 24 categories**, enabling on-demand evaluation and training beyond manually curated task sets.
- **MOSSBench: Is Your Multimodal LLM Oversensitive to Safe Queries?** [ICLR 2025] [Code]
  The first benchmark for multimodal oversensitivity.
  Introduced **300 human-verified benign multimodal queries** across three visual-trigger categories to measure unjustified safety refusals. Evaluated **20 MLLMs**, finding refusal rates up to **76%** and tracing failures to perception, intent reasoning, and safety judgment.

Research Experience

- **Amazon Web Services (AWS)** — *Jun 2026 – Present*
  Applied Research Intern | AgentCore — New York City, USA
  ◦ **Topic:** Multimodal coding agents for 3D scene reconstruction. **Paper:** LEGO-Anything.
- **University of Maryland, College Park** — *Sep 2025 – Present*
  Research Assistant | Multimodal Agents & Agent Environments — College Park, USA
  ◦ **Topics:** Embodied agents, agent environments, and visual reasoning. **Papers:** Guava, ClawEnvKit, Frankenstein.
- **University of California, Los Angeles** — *Aug 2024 – Aug 2025*
  Research Assistant | Multimodal Learning & Safety — Los Angeles, USA
  ◦ **Topics:** RL for visual reasoning, multimodal safety, and LLM red-teaming. **Papers:** VisualThinker, MOSSBench, DrAttack.

Skills & Service

- **Programming:** Python, C++, SQL, Java, Shell, MATLAB.
- **ML & Infrastructure:** PyTorch, VERL, Ray, Hugging Face, DeepSpeed, AWS, Docker, Git, Linux.
- **Domains:** Multimodal agents, visual reasoning, reinforcement learning, embodied AI, agent environment design.
- **Service:** Reviewer at ICML, NeurIPS, COLM, ICLR (2026).